

1.5 Part - A : Two Marks Questions

Question 1. Form the partial differential equation by eliminating the constants a and b from $z = ax + by + ab$.

Solution . Given equation $z = ax + by + ab$.

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = a. \Rightarrow a = p.$$

$$\frac{\partial z}{\partial y} = q = b. \Rightarrow b = q$$

Using (2) and (3) in (1), we get $z = px + qy + pq$.

Question 2. Form the partial differential equation by eliminating the constants from $z = ax + by + a^2 + b^2$.

Solution . Given equation $z = ax + by + a^2 + b^2$.

Differentiate (1) partially w.r.to x and y , we get

$$\frac{\partial z}{\partial x} = p = a. \Rightarrow a = p.$$

$$\frac{\partial z}{\partial y} = q = b. \Rightarrow b = q$$

Using (2) and (3) in (1), we get $z = px + qy + p^2 + q^2$.

Question 3. Form the partial differential equation by eliminating arbitrary constants from $(x - a)^2 + (y - b)^2 + z^2 = c^2$. Where c is a fixed constant.

Solution . Given equation $(x - a)^2 + (y - b)^2 + z^2 = c^2$.

Differentiate (1) partially w.r.to x and y , we get

$$2(x - a) + 2z.p = 0. \Rightarrow (x - a) = -zp.$$

$$2(y - b) + 2z.q = 0. \Rightarrow (y - b) = -zq.$$

Using (2) and (3) in (1), we get

$$\begin{aligned} (-zp)^2 + (-zq)^2 + z^2 &= c^2. \\ \Rightarrow z^2(p^2 + q^2 + 1) &= c^2. \end{aligned}$$

Question 4. Form the partial differential equation by eliminating the constants from $(x - a)^2 + (y - b)^2 + z^2 = 1$. Nov.'12

Solution . Given equation $(x - a)^2 + (y - b)^2 + z^2 = 1$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$2(x - a) + 2z.p = 0. \Rightarrow (x - a) = -zp. \quad \dots(2)$$

$$2(y - b) + 2z.q = 0. \Rightarrow (y - b) = -zq. \quad \dots(3)$$

Using (2) and (3) in (1), we get

$$\begin{aligned} (-zp)^2 + (-zq)^2 + z^2 &= 1. \\ \Rightarrow z^2(p^2 + q^2 + 1) &= 1. \end{aligned}$$

Question 5. Form the partial differential equation by eliminating the constants a and b from $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$. Apr.'13

Solution . Given equation $(x - a)^2 + (y - b)^2 = z^2 \cot^2 \alpha$ (1)

Differentiate (1) partially w.r.to x and y , we get

$$2(x - a) = 2z \cot^2 \alpha \cdot \frac{\partial z}{\partial x}, \Rightarrow (x - a) = zp \cot^2 \alpha. \quad \dots(2)$$

$$2(y - b) = 2z \cot^2 \alpha \cdot \frac{\partial z}{\partial y}, \Rightarrow (y - b) = zq \cot^2 \alpha. \quad \dots(3)$$

Substitute (2) and (3) in (1), we get

$$\begin{aligned} z^2 p^2 \cot^4 \alpha + z^2 q^2 \cot^4 \alpha &= z^2 \cot^2 \alpha. \\ \Rightarrow p^2 + q^2 &= \tan^2 \alpha. \end{aligned}$$

Question 12. Form the partial differential equation by eliminating the arbitrary function 'f' from $f(z - xy, x^2 + y^2)$.

Nov.'10, Nov.'11, Apr.'14

Solution . Let the given function is of the form $f(u, v) = 0$.

Where $u = z - xy, v = x^2 + y^2$.

Then $\frac{\partial u}{\partial x} = p - y$ and $\frac{\partial u}{\partial y} = q - x$ Also $\frac{\partial v}{\partial x} = 2x$ and $\frac{\partial v}{\partial y} = 2y$

Thus the elimination of f is,
$$\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} & \frac{\partial v}{\partial y} \end{vmatrix} = 0$$

i.e.,
$$\begin{vmatrix} p - y & 2x \\ q - x & 2y \end{vmatrix} = 0$$

$$2y(p - y) - 2x(q - x) = 0$$

$$y(p - y) - x(q - x) = 0$$

$$yp - xq = y^2 - x^2.$$

Question 13. Find the complete integral of $p + q = pq$

Nov.'11

Solution . Given equation is $p + q = pq$

It is of the form $F(p, q) = 0$.

Let the solution of (1) be $z = ax + by + c$.

Partially differentiate (2) w.r.to x and y we get $p = a$ and $q = b$

Substitute $p = a$ and $q = b$ in (1) we get

$$a + b = ab, \text{ i.e., } ab - b = a. \Rightarrow b = \frac{a}{a-1}$$

Substitute b value in (2) we get $z = ax + \frac{a}{a-1}y + c$.

Which is the complete integral.

Question 14. Solve : $xp + yq = z$.

Nov.'10, Apr.'14

Solution . Given equation is $xp + yq = z$.

This is Lagrange's type $Pp + Qq = R$. The auxiliary equation is,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z} \quad \dots(2)$$

Where $P = x$, $Q = y$ and $R = z$.

Taking first and second ratio, we have

$$\frac{dx}{x} = \frac{dy}{y}$$

Integrating, $\int \frac{dx}{x} = \int \frac{dy}{y} + \log c_1$

$$\log x = \log y + \log c_1$$

$$\log \frac{x}{y} = \log c_1$$

$$\Rightarrow \frac{x}{y} = c_1. \quad \dots(3)$$

Taking second and third ratio, we have

$$\frac{dy}{y} = \frac{dz}{z}$$

Integrating, $\int \frac{dy}{y} = \int \frac{dz}{z} + \log c_2$

$$\log y = \log z + \log c_2$$

$$\log \frac{y}{z} = \log c_2$$

$$\Rightarrow \frac{y}{z} = c_2. \quad \dots(4)$$

From (3) and (4), the general solution is $\phi\left(\frac{x}{y}, \frac{y}{z}\right) = 0$.

Question 15. Find complete solution of $(p + q)(z - px - qy) = 1$.

Apr.'12

Solution . Given equation $(p + q)(z - px - qy) = 1$ is written as

$$(z - px - qy) = \frac{1}{p + q}$$

$$\therefore z = px + qy + \frac{1}{p + q} \quad \dots(1)$$

This is of the Clairaut's form $z = px + qy + f(p, q)$

∴ The complete integral is (Replace p by a and q by b in (1))

$$z = ax + by + \frac{1}{a+b}$$

Question 16. Solve $(D^2 - 2DD' + D'^2)z = 0$.

Apr.'11

Solution . Given equation is, $(D^2 - 2DD' + D'^2)z = 0$.

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^2 - 2m + 1 = 0$$

$$(m - 1)^2 = 0 \Rightarrow m = 1, 1$$

The required solution is, $z = f_1(y + x) + xf_2(y + x)$.

Question 17. Solve $(D^2 - 4DD' + 4D'^2)z = 0$.

Nov.'11

Solution . Given equation is, $(D^2 - 4DD' + 4D'^2)z = 0$.

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^2 - 4m + 4 = 0$$

$$(m - 2)^2 = 0 \Rightarrow m = 2, 2$$

The required solution is, $z = f_1(y + 2x) + xf_2(y + 2x)$.

Question 18. Solve $(D^3 - 3DD'^2 + 2D'^3)z = 0$. Nov.'12, Apr.'13

Solution . Given equation is, $(D^3 - 3DD'^2 + 2D'^3)z = 0$.

Replace D by m and D' in (1). Then the auxiliary equation is

$$m^3 - 3m + 2 = 0$$

$$(m - 1)(m^2 + m - 2) = 0$$

$$(m - 1)(m + 2)(m - 1) = 0 \Rightarrow m = -1, -1, 2$$

The required solution is,

$$z = f_1(y - x) + xf_2(y - x) + f_3(y + 2x)$$

Question 19. Solve $(D^2 - 6DD' + 9D'^2)z = 0$.

Nov.'14

Solution . Given equation is, $(D^2 - 6DD' + 9D'^2)z = 0$ (1)

Replace D by m and D' in (1). Then the auxiliary equation is

$$\begin{aligned} m^2 - 6m + 9 &= 0 \\ (m - 3)^2 &= 0. \Rightarrow m = 3, 3. \end{aligned}$$

The required solution is,

$$z = f_1(y + 3x) + x f_2(y + 3x).$$

Question 20. Solve : $2\frac{\partial^2 z}{\partial x^2} - 5\frac{\partial^2 z}{\partial x \partial y} + 2\frac{\partial^2 z}{\partial y^2} = 0$

Apr.'15

Solution . Given equation is written as, $(2D^2 - 5DD' + 2D'^2)z = 0$.

Auxiliary equation is [Replace D by m and D']

$$\begin{aligned} 2m^2 - 5m + 2 &= 0 \\ 2m^2 - 4m - m + 2 &= 0 \\ 2m(m - 2) - 1(m - 2) &= 0. \\ (2m - 1)(m - 2) &= 0. \Rightarrow m = \frac{1}{2}, 2 \end{aligned}$$

The required solution is,

$$z = f_1\left(y + \frac{1}{2}x\right) + f_2(y + 2x).$$

Question 21. Solve : $(D - D')^3 z = 0$.

Nov.'15

Solution . Replace $D = m$ and $D' = 1$ then the auxiliary equation is $(m - 1)^3 = 0. \Rightarrow m = 1, 1, 1$.

Thus the solution is $y = f_1(y + x) + x f_2(y + x) + x^2 f_3(y + x)$.

Question 22. Find the particular integral of

$$(D^2 + 2DD' + D'^2)z = e^{x-y}.$$

Apr.'12

Solution .

$$P.I. = \frac{1}{D^2 + 2DD' + D'^2} e^{x-y}$$

Replace D by 1 and D' by -1 we get

$$= \frac{1}{1 - 2 + 1} e^{x-y} = \frac{1}{0} e^{x-y}$$

$$= \frac{x}{2D - 2D'} e^{x-y} = \frac{x}{2 + 2} e^{x-y}$$

$$\therefore P.I. = \frac{x}{4} e^{x-y}$$

Question 23. Find the particular integral of $(D^2 + 4DD')z = e^x$
Solution . Nov. '13

$$P.I. = \frac{1}{D^2 + 4DD'} e^{x+0y}$$

Replace D by 1 and D' by 0 we get

$$= \frac{1}{1^2 + 4 \cdot 0} e^{x+0y} = e^x$$

Question 24. Find the PI of $(D^2 + 3DD' + 2D'^2)z = \sin(x + 5y)$.
Solution . Nov. '15

$$P.I. = \frac{1}{D^2 + 3DD' + 2D'^2} \sin(x + 5y)$$

Replace $D^2 = -1^2$ $DD' = -1 \times 5$ and $D'^2 = -5^2$ we get

$$= \frac{1}{1 + 3(-5) + 2(-25)} \sin(x + 5y)$$

$$\therefore P.I. = -\frac{1}{64} \sin(x + 5y).$$

Question 25. Find the PI of $r - 4s + 4t = e^{2x+y}$.
Solution .

$$P.I. = \frac{1}{D^2 - 4DD' + 4D'^2} e^{2x+y} \quad \text{Replace } D = 2 \text{ and } D' = 1$$

$$= \frac{1}{4 - 8 + 4} e^{2x+y} = \frac{1}{0} e^{2x+y}$$

$$= \frac{x}{2D - 4D'} e^{2x+y} = \frac{x}{4 - 4} e^{2x+y} = \frac{x}{0} e^{2x+y}$$

$$\therefore P.I. = \frac{x^2}{2} e^{2x+y}.$$